



Pricing bonds with make-whole call provisions

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Introduction to Make-Whole Bonds

- ▶ A **Make-Whole Bond** is a special type of callable bond whose redemption price is calculated based on a market benchmark interest rate plus a predetermined spread.
- ▶ The system implements the Make-Whole pricing model described in **Lin Ma et al. (2023)**.
- ▶ Purpose: disincentivize opportunistic refinancing by the issuer while fully compensating the bondholder for the loss of investment.
- ▶ The issuer can call the bond but must **”make the investor whole”** by paying a price that covers the present value of all remaining cash flows, discounted at a current market benchmark rate plus a contractual penalty spread.

Make-Whole Call Strike Price Formula

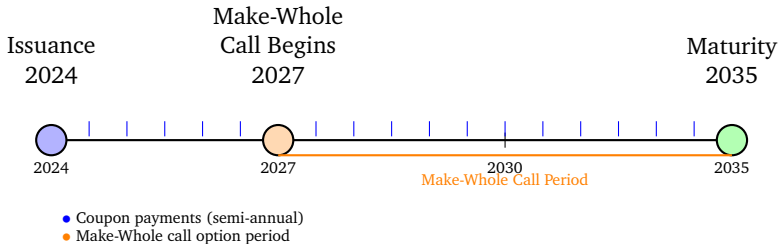
According to Lin Ma et al. (2023), the Make-Whole Call strike price for bond i on a potential call date t is:

$$P_{i,t}^{\text{make-whole call}} = \sum_{n=1}^N \frac{CF_{i,t+n}}{(1 + \text{swaprate}_{i,t+n} + \text{makewholepremium}_i)^n}$$

where:

- ▶ $CF_{i,t+n}$: future cash flow (coupon and principal) at time $t + n$.
- ▶ $\text{swaprate}_{i,t+n}$: benchmark interest rate (swap rate) for maturity n at time t .
- ▶ $\text{makewholepremium}_i$: contractual spread added to the benchmark rate (the “make-whole” penalty).

Make-Whole Bond Timeline



Interpretation of the Formula

- ▶ The formula is a **sum of discounted cash flows**, a standard valuation technique.
- ▶ The discount rate for each cash flow is the **maturity-matched benchmark rate plus a fixed penalty spread**.
- ▶ Using maturity-matched rates accounts for the **shape of the yield curve** more accurately than a single yield.
- ▶ When market rates fall, the discount rates (swap rates) decrease, but the fixed premium mitigates the decrease slightly.
- ▶ The resulting strike price is generally at a **premium** to the bond's face value and market price, compensating the holder for the loss of above-market coupon stream.
- ▶ Make-whole calls are rarely exercised purely for interest-rate savings; typically invoked for structural reasons (mergers, asset sales).

Data Sources

- ▶ Bond data: **WRDS Mergent FISD database.**
- ▶ Treasury rate curve: **U.S. Department of the Treasury.**

Bond Identifiers (Linking Keys)

Variable	Role in Strike Price Construction
issueid	Primary key for linking all bond-related information.
issuerid	Links to issuer-level information.
completecusip	Unique security identifier for validation and external data linking.
issuecusip	Alternative security identifier.

Make-Whole Premium (Spread)

- ▶ **makewholespread**: the spread above a comparable Treasury issue that will have to be paid if the make-whole call option is exercised.
- ▶ Direct source for $\text{makewholepremium}_i$; textual field (e.g., “0.50%”) must be parsed into a numeric value.
- ▶ **investmentgrade**: flag (Y/N) used as a fallback rule: if **makewholespread** is missing, assume 25bps for investment-grade and 50bps for high-yield bonds.

Bond Type Determination and Call Schedule

Variable	Role in Strike Price Construction
makewholestartdate	Defines the first possible call date t for the Make-Whole option.
makewholeenddate	For a pure Make-Whole bond, this is the last possible call date; helps define the period $[t, t + N]$.
nextcalldate	For a hybrid bond, this is the date $t + N$ (the first fixed-price call date); cash flows are calculated up to this date.
nextcallprice	For a hybrid bond, this is the final cash flow $CF_{i,t+N}$ (converted to dollar amount).
maturity	For a pure Make-Whole bond, this is the date $t + N$; the final cash flow is the principal plus final coupon.
callable	Flag used to filter bonds that are eligible for a Make-Whole call calculation.

Cash Flow (CF) Calculation

Variable	Role in Strike Price Construction
principalamt	Used to calculate the dollar amount of coupon payments and the final principal repayment.
coupon	The annual coupon rate used to calculate periodic coupon payments.
interestfrequency	Determines the number of coupon payments per year and the timing of cash flows.
daycountbasis	Used for precise calculation of accrued interest and coupon amount for each period.
dateddate	Reference point for calculating the first coupon period.
firstinterestdate	Used to generate the schedule of coupon payment dates.
lastinterestdate	Can be used to verify the coupon schedule near maturity.

Treasury Rate Processing in Make-Whole Pricing

- ▶ **Treasury Curve Provider:** Loads and manages U.S. Treasury par yield curve data (1-month to 30-year maturities) for the benchmark interest rates required by the formula.
- ▶ **Data preprocessing:**
 - ▶ Percentage to decimal conversion (e.g., 7.83% $\rightarrow$ 0.0783)
 - ▶ Forward-fill missing values; create maturity mapping (e.g., 1mo $\rightarrow$ 0.0833years, 30yr $\rightarrow$ 30years)
- ▶ **Maturity-matched interpolation:**
 - ▶ For a cash flow occurring n periods after the call date, compute its exact term in years:
$$\text{years_from_start} = \frac{\text{cash flow date} - \text{call date}}{365.25}$$
 - ▶ Interpolate the Treasury curve at that exact term using linear, cubic, or other methods to obtain $\text{swaprate}_{i,t+n}$

Treasury Rate Processing in Make-Whole Pricing

- ▶ **Application in the discounting formula:**
 - ▶ Each cash flow $CF_{i,t+n}$ is discounted at discount rate $rate_{t+n} = treasury_rate_{t+n} + makewholepremium_i$
 - ▶ The period discount rate is scaled by the period length:
 $period_discount_rate = annual_discount_rate \times period_years$
 - ▶ Discount factor: $\frac{1}{(1+period_discount_rate)^n}$
- ▶ **Weighted-average Treasury rate** (for spread calculation):
 - ▶ $\bar{r}_{treasury} = \frac{\sum_{n=1}^N r_{t+n} \cdot PV(CF_{t+n})}{\sum_{n=1}^N PV(CF_{t+n})}$
 - ▶ Spread = Coupon Rate - $(\bar{r}_{treasury} + makewholepremium_i)$
- ▶ **Robust date handling:** If the call date is not a trading day, the curve of the nearest preceding trading day is used automatically.

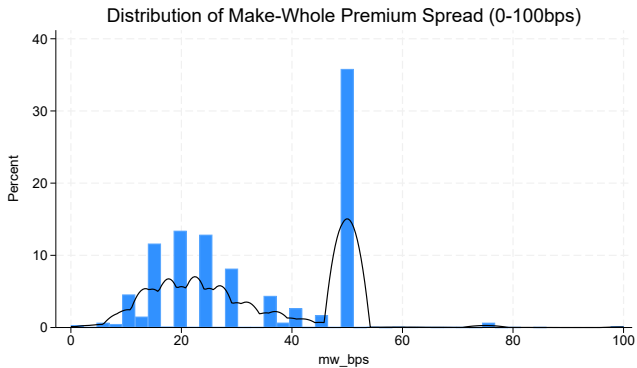
Constructing Future Cash Flows (CF)

- ▶ First, determine **bond type** (pure vs. hybrid) to define the endpoint $t + N$ and the final payment.
 - ▶ **Pure Make-Whole bond**: makewholeenddate $\approx$ maturity; cash flows continue until legal maturity; final payment = principal + last coupon.
 - ▶ **Hybrid bond**: makewholeenddate $<$ maturity; cash flows end at the first fixed-price call date; final payment = fixed call price $\times$ principal.
- ▶ Identify all coupon payment dates within the period using:
 - ▶ Annual coupon rate, payment frequency, first interest date, maturity date, day-count convention.
- ▶ Calculate cash-flow amounts:
 - ▶ Coupon payments: principal $\times$ coupon rate $\times$ day-count fraction.
 - ▶ Final payment: according to bond type.
- ▶ Pair each future time point $t + n$ with its precise cash amount $CF_{i,t+n}$ to form the complete cash-flow sequence.

Sample Construction

- ▶ All bonds with a make-whole call provision from Mergent FISD: **32,455 bonds**.
- ▶ **Coupon rate:**
 - ▶ Mean: 5.37%, Std. dev.: 3.11, Median: 5.15%.
 - ▶ Strong right-skew (skewness = 6.22) and leptokurtic (kurtosis = 142.14).
 - ▶ Range: 0% to 99.876% (extreme outlier).
 - ▶ Central 50%: 3.75% to 6.625%.
 - ▶ 90% of bonds have coupon $\leq$ 8.625%; only 1% exceed 12%.
- ▶ **Make-whole spread:**
 - ▶ 24,379 observations; mean 33.24bps, std. dev. = 15.81.
 - ▶ Positively skewed (skewness = 0.73), leptokurtic (kurtosis = 12.32).
 - ▶ Majority between 20bps (25th percentile) and 50bps (75th percentile).
 - ▶ 90% of observations have spread $\leq$ 50bps; maximum observed: 375bps.
 - ▶ Missing values replaced with 50bps (median).

Sample Construction



Calculation Results for Pure Make-Whole Bonds

- ▶ **20,684** pure make-whole bonds.
- ▶ Strike price per \$100 face value (price per 100):
 - ▶ Mean: 117.547, Std. dev.: 15.76789.
 - ▶ Median: 114.0078.
 - ▶ 10th percentile: 100.81, 95th percentile: 146.38.
 - ▶ 1% of bond prices exceed 162.63.
- ▶ Extreme values present (smallest: 56.5564, largest: 234.8361) confirm heavy-tailed distribution.
- ▶ Example: CUSIP 65504L (coupon 8.95%, 30-year term) in a low-interest-rate environment (2.67%) yields a large annualized spread, amplifying the present value.
- ▶ **Key insight:** Call timing has a huge impact on bond prices; interest rates affect the call price, causing fluctuations.
- ▶ The Make-Whole provision protects investors by ensuring the call price covers reinvestment risk.

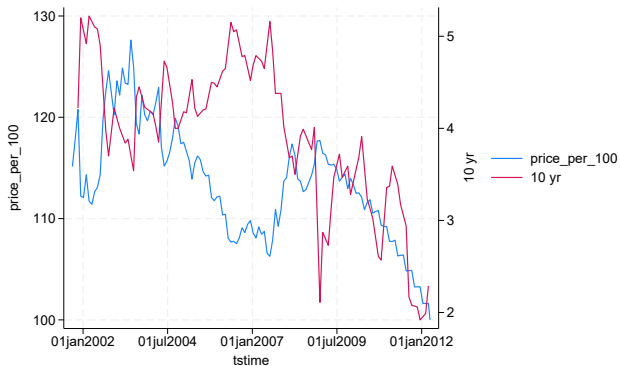
Make-Whole Treasury Spread Distribution (Pure)

- ▶ Spread = Coupon Rate – (Weighted Average Treasury Rate + MW Premium).
- ▶ Descriptive statistics:
 - ▶ Mean: 0.03020, Median: 0.02320, Std. dev.: 0.02432.
 - ▶ Right-skewed (skewness = 1.12939), leptokurtic (kurtosis = 4.04471).
 - ▶ 1st percentile: –0.00537, 99th percentile: 0.10201.
 - ▶ Minimum: –0.03800, Maximum: 0.17316.
- ▶ Interpretation:
 - ▶ Right-skew indicates most bonds have moderate excess returns, but a “high-risk, high-return” tail exists (likely lower-rated high-yield bonds).
 - ▶ Negative spreads imply the coupon rate is below the risk-free benchmark (“deep out-of-the-money”).
 - ▶ High spreads reflect substantial risk premiums demanded by investors.

Make-Whole Bond Strike Price Curves

- ▶ Due to limited database access, the strike price is calculated monthly from the make-whole start date to the make-whole end date.
- ▶ Curves show how make-whole bond prices are influenced by fluctuations in Treasury rates.
- ▶ **Key observation:** Bond prices exhibit a strict negative correlation with Treasury rates, forming an almost perfect inverse mirror image.
- ▶ Historical phases:
 1. **2002–2004 (rate-cutting cycle):** Declining Treasury yields pushed make-whole bond prices to historical highs (duration effect).
 2. **2004–2007 (rate-hiking cycle):** Prices trended downward as interest rates rose.
 3. **Post-2007 (financial crisis):** Despite sharp drop in interest rates, prices failed to reach new highs; time-decay effect dominates as maturity approaches, forcing convergence toward par value.

Make-Whole Bond Strike Price Curves



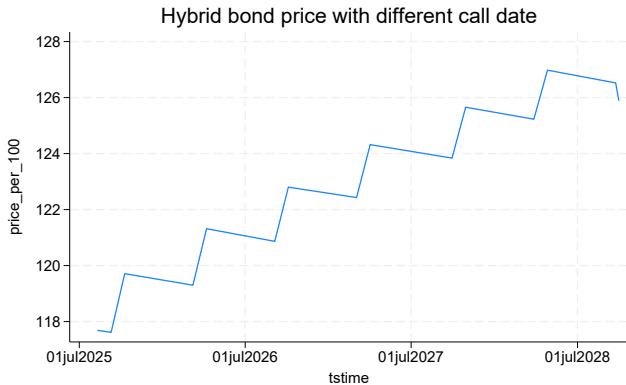
Make-Whole Bond Strike Price Curves



Hybrid Bond Pricing

- ▶ Hybrid bonds combine two redemption mechanisms:
 1. Early call period: issuer can redeem at a Make-Whole price (market rates + fixed premium).
 2. After a specified date: redemption at a predetermined fixed price (typically declining over time).
- ▶ Provides issuers with financing flexibility and a known cost ceiling.
- ▶ For investors: introduces dual uncertainty (timing and method of redemption), increasing uncertainty about holding period and final yield.
- ▶ Investors typically demand an additional yield premium (approx. 10–50bps) as compensation.
- ▶ Pricing model: Make-Whole price calculation is limited to the period before the first fixed-price call date, and includes the fixed call price at that date.

Hybrid Bond Pricing



The call dates are generated by next_call_date and the last date is the maturity

Conclusion

- ▶ The Make-Whole bond pricing system successfully implements the model of LinMaetal. (2023).
- ▶ The strike-price formula elegantly translates contractual protection into a precise, market-based mathematical price.
- ▶ Results show that make-whole bonds provide effective protection for investors against reinvestment risk in declining-rate environments.
- ▶ The distribution of make-whole spreads and strike prices reveals the risk structure and pricing logic of the market.
- ▶ Hybrid bonds introduce additional complexity, offering flexibility for issuers but requiring higher compensation for investors.
- ▶ Future work: obtain complete call-schedule data for hybrid bonds to enable full pricing validation.

Thank you for your attention!